Hypothesis Testing & Confidence Intervals

🎯 Session Objectives:

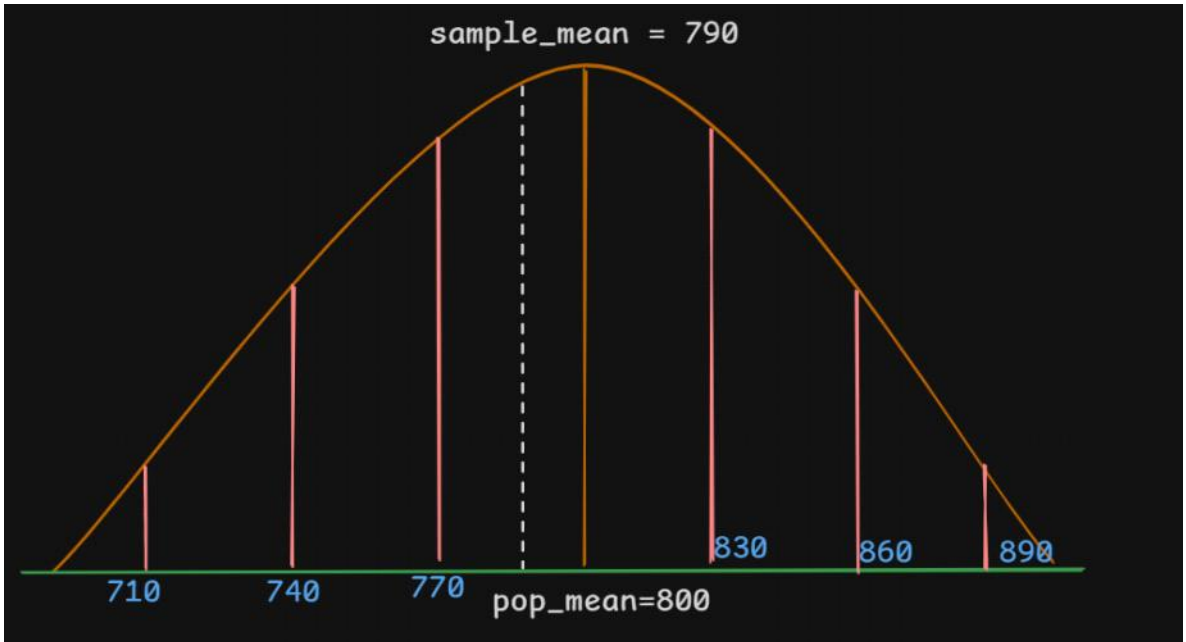
- ✅ Understand and articulate the steps involved in hypothesis testing.
- ✅ Formulate null and alternate hypotheses for various scenarios.
- ✅ Explain the role of normal and standard normal distributions in hypothesis testing.
- ✅ Define and interpret p-values and significance levels (alpha) in the context of hypothesis testing.
- ✅ Construct appropriate null and alternate hypotheses based on different practical situations.
- ✅ Understand the concept of confidence intervals and how they relate to hypothesis testing.
- ✅ Construct confidence intervals using Python with detailed examples.
- ✅ Gain a basic understanding of Type I and Type II errors, relating them to the concepts of false positives and false negatives.

Z	.00	.01	.02	.03	.04	.05	.06
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914

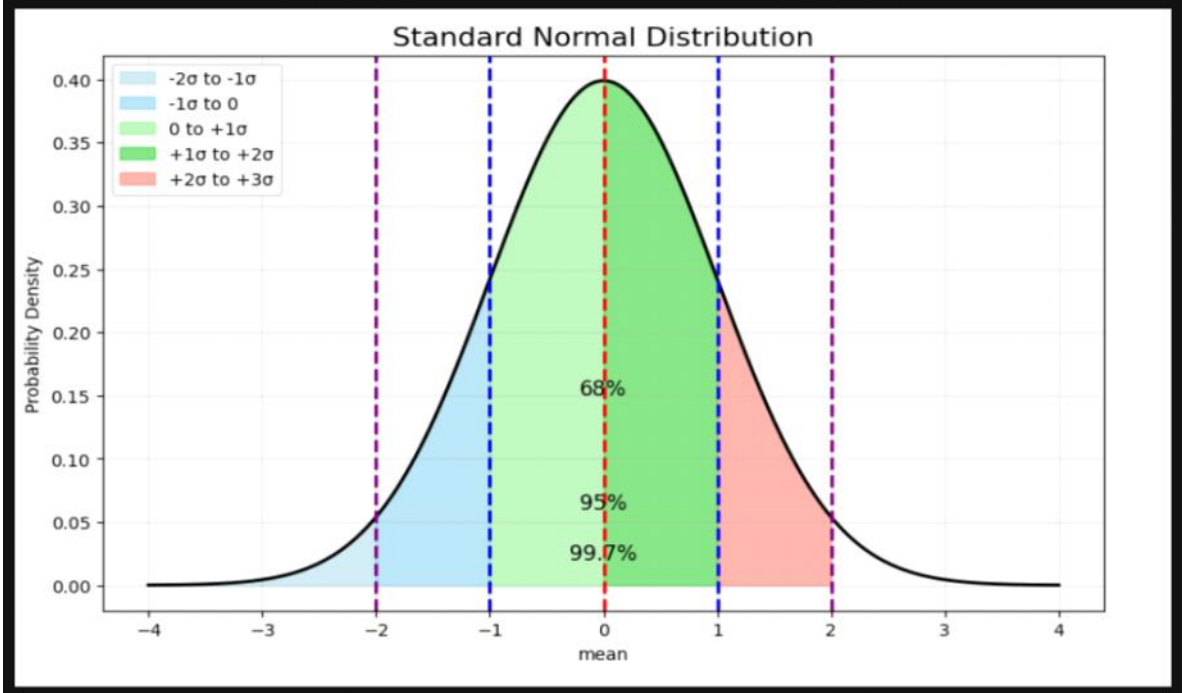
```python
# 2-Tailed Test [Alternative Hypothesis can be +ve or -ve]
p_value = 2 * norm.cdf(z_score)
p_value
```

```
0.018422125454099002
```

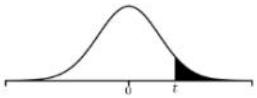
```python
# Light Bulb Lifespan
population_mean = 800
sample_mean = 790
sample_size = 50
sample_std = 30
significance_level = 0.05
```



-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383



Critical Values for Student's t-Distribution.



df	Upper Tail Probability: $\Pr(T > t)$ 0.2	0.1	0.05	0.04	0.03	0.025	0.02	0.01	0.005	0.0005
1	1.376	3.078	6.314	7.916	10.579	12.706	15.895	31.821	63.657	636.619
2	1.061	1.886	2.920	3.320	3.896	4.303	4.849	6.965	9.925	31.599
3	0.978	1.638	2.353	2.605	2.951	3.182	3.482	4.541	5.841	12.924
4	0.941	1.533	2.132	2.333	2.601	2.776	2.999	3.747	4.604	8.610
5	0.920	1.476	2.015	2.191	2.422	2.571	2.757	3.365	4.032	6.869
6	0.906	1.440	1.943	2.104	2.313	2.447	2.612	3.143	3.707	5.959
7	0.896	1.415	1.895	2.046	2.241	2.365	2.517	2.998	3.499	5.408
8	0.889	1.397	1.860	2.004	2.189	2.306	2.449	2.896	3.355	5.041
9	0.883	1.383	1.833	1.973	2.150	2.262	2.398	2.821	3.250	4.781
10	0.879	1.372	1.812	1.948	2.120	2.228	2.359	2.764	3.169	4.587

3

Standard Error of Proportion

$SE = \sqrt{[p(1-p)/n]}$ measures variability of sample proportion

02

Calculate Standard Error

$SE = \sqrt{[p_pool \times (1 - p_pool) \times (1/n_1 + 1/n_2)]} \approx 0.0142$

Z	.00	.01	.02	.03	.04	.05
0.0	.50000	.50399	.50798	.51197	.51595	.51994
0.1	.53983	.54380	.54776	.55172	.55567	.55962
0.2	.57926	.58317	.58706	.59095	.59483	.59871
0.3	.61791	.62172	.62552	.62930	.63307	.63683
0.4	.65542	.65910	.66276	.66640	.67003	.67364
0.5	.69146	.69497	.69847	.70194	.70540	.70884
0.6	.72575	.72907	.73237	.73565	.73891	.74215
0.7	.75804	.76115	.76424	.76730	.77035	.77337

```
z_score = (sample_mean - population_mean) / StandardError
z_score
```

```
0.7050905410411598
```

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062

```
1-0.881 = 0.119
```

Standard Error

SE = 1.2 / √50 ≈ 0.170

Z-Score

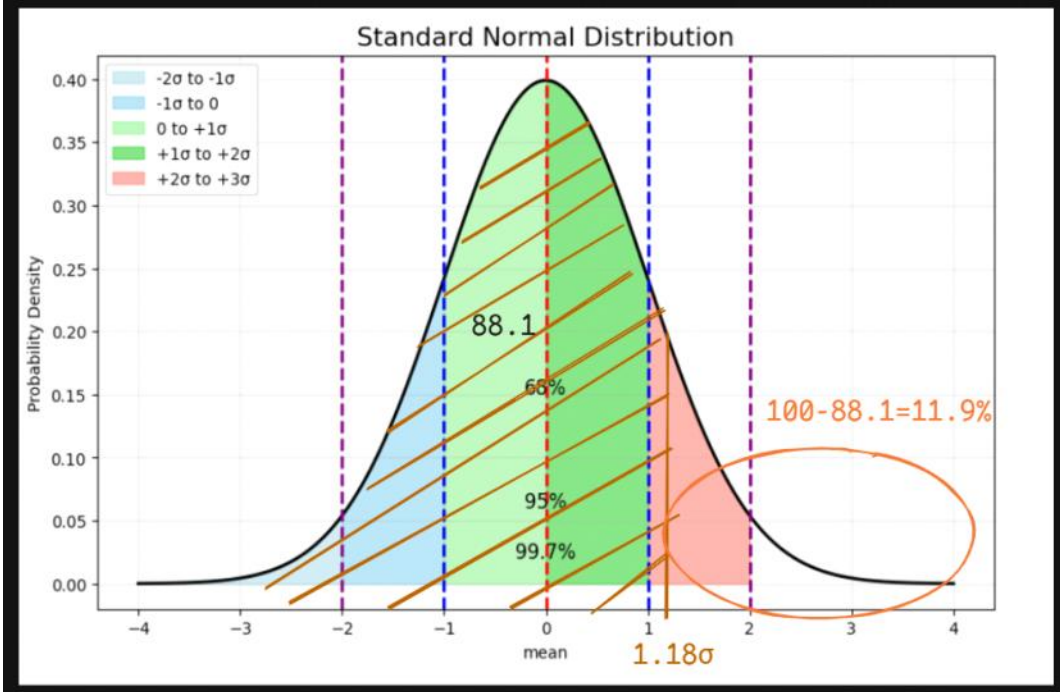
z = (5.7 - 5.5) / 0.170 ≈ 1.179

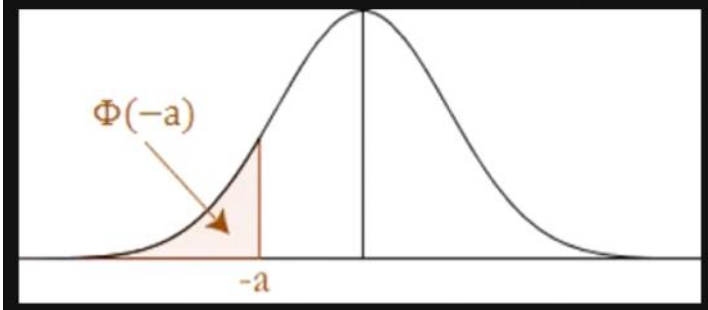
P-Value

One-tailed: ≈ 0.119

Decision

Fail to reject H_0





Impact of Sample Size on Confidence Intervals

Larger Sample Size

When sample size increases, the standard error decreases. A smaller standard error results in a **narrower confidence interval**, providing a more precise estimate of the population parameter.

Example: A sample of 100 flour packages yields a narrower 95% CI compared to 25 packages, because the sample mean is more reliable.

Smaller Sample Size

A smaller sample size increases the standard error, leading to a **wider confidence interval**. This means our estimate is less precise, and the range is broader.

Example: Sampling only 10 flour packages produces a wider 95% CI due to greater uncertainty from the smaller sample.

Confidence Level	Z-Score (Two-tailed)	% in Each Tail
80%	1.28	10%
85%	1.44	7.5%
90%	1.645	5%
92%	1.75	4%
95%	1.96	2.5%
96%	2.05	2%
97%	2.17	1.5%
98%	2.33	1%
99%	2.575	0.5%
99.5%	2.807	0.25%
99.9%	3.291	0.05%

Example 1: Mean Weight of Packaged Flour

Scenario

A company claims the average weight of its packaged flour is 1.0 kg. You sample 40 packages and find an average weight of 0.98 kg with a standard deviation of 0.1 kg. Construct a 95% confidence interval for the true mean weight.

01

Define Known Parameters

Sample mean = 0.98 kg, Sample size = 40, Standard deviation = 0.1 kg, Confidence level = 95%

02

Calculate Standard Error

$SE = s / \sqrt{n} = 0.1 / \sqrt{40} = 0.0158$ kg

03

Find Z-Score

For 95% confidence interval, Z-score = 1.96

04

Construct Confidence Interval

$CI = \bar{x} \pm (Z \times SE) = 0.98 \pm (1.96 \times 0.0158)$

```
# Hypothesis Testing
from scipy.stats import norm
import numpy as np

# Light Bulb Lifespan
population_mean = 800
sample_mean = 790
sample_size = 50
sample_std = 30
significance_level = 0.05

Standard_Error = sample_std / np.sqrt(sample_size)
Standard_Error
```

```
4.242640687119285
```

```
z_score = (sample_mean - population_mean) / Standard_Error
z_score
```

```
-2.3570226039551585
```

```
H0 = "The lifespan is not significantly different from 800 hours."
HA = "The lifespan is significantly different from 800 hours. " # != 800
# 2-Tailed Test [Alternative Hypothesis can be +ve or -ve]
p_value = 2 * norm.cdf(z_score)
p_value
```

```
0.018422125454099002
```

```
alpha = 0.05 # Significance Level
if (p_value < alpha):
    print(f"Rejecting the Null Hypothesis H0: {H0}")
    print(f"Accepting the Alternative Hypothesis HA: {HA}")
else:
    print(f"Accepting the Null Hypothesis H0: {H0}")
```

```
Rejecting the Null Hypothesis H0: The lifespan is not significantly different from 800 hours.
Accepting the Alternative Hypothesis HA: The lifespan is significantly different from 800 hours.
```

Conclusion: The average lifespan of the light bulbs is significantly different from 800 hours. The factory's claim is not supported by the sample data.

```
# Hypothesis Testing
from scipy.stats import norm
import numpy as np

# Flour Package Weight [Another Example] - with 40 sample size and 0.1 std
population_mean = 1.0 # in kgs
sample_mean = 0.98 # in kgs
sample_size = 40 # n>=30 : Z-Score
sample_std = 0.10 # is known to me

Standard_Error = sample_std / np.sqrt(sample_size)
Standard_Error
```

```
0.015811388300841896
```

```
z_score = (sample_mean - population_mean) / Standard_Error
z_score
```

```
-1.2649110640673529
```

```
H0 = "The average weight of packaged flour is 1 kg"
HA = "The average weight of packaged flour is not 1 kg" # != 1
# 2-Tailed Test [Alternative Hypothesis can be +ve or -ve]
p_value = 2 * norm.cdf(z_score)
p_value
```

```
0.20590321073206785
```

```
alpha = 0.05 # Significance Level
if (p_value < alpha):
    print(f"Rejecting the Null Hypothesis H0: {H0}")
    print(f"Accepting the Alternative Hypothesis HA: {HA}")
else:
    print(f"Accepting the Null Hypothesis H0: {H0}")
```

```
Accepting the Null Hypothesis H0: The average weight of packaged flour is 1 kg
```

Conclusion: Since the p-value (0.2059) is greater than 0.05, we do not reject the null hypothesis. The average weight is not significantly different from 1.0 kg.

```python
# A/B Testing [Hypothesis Testing]
from scipy.stats import norm
import numpy as np

# A company runs an A/B test on their website to see if a new design
# (Version B) improves conversion rate compared to the old design (Version A).

population_mean = 120/1000 # Version A
sample_mean = 156/1200 # Version B

# sample size
n1 = 1000
n2 = 1200

p_pool = (120 + 156) / (1000 + 1200)
p_pool
```

```
0.12545454545454546
```

```python
StandardError = np.sqrt(p_pool * (1-p_pool) * (1/n1+1/n2))
StandardError
```

```
0.014182575737342443
```

```python
z_score = (sample_mean - population_mean) / StandardError
z_score
```

```
0.7050905410411598
```

```python
H0 = "No Significant Difference Exist between the designs."
HA = "The new design has an effect"
# 1 Tailed Testing
p_value = norm.cdf(z_score)
p_value
```

```
0.759623053685014
```

```python
alpha = 0.05 # Significance Level
if (p_value < alpha):
    print(f"Rejecting the Null Hypothesis H0: {H0}")
    print(f"Accepting the Alternative Hypothesis HA: {HA}")
else:
    print(f"Accepting the Null Hypothesis H0: {H0}")
```

```
Accepting the Null Hypothesis H0: No Significant Difference Exist between the designs.
```

Conclusion: There is no significant difference in conversion rates between the two website designs. The new design does not appear to have improved the conversion rate.

```python
# Mobile App Session Duration
from scipy.stats import norm
import numpy as np

population_mean = 5.5 # in kgs
sample_mean = 5.7 # in kgs
sample_size = 50 # n>=30 : Z-Score
sample_std = 1.2 # is known to me

Standard_Error = sample_std / np.sqrt(sample_size)
Standard_Error
```

```
0.1697056274847714
```

```python
z_score = (sample_mean - population_mean) / Standard_Error
z_score
```

```
1.1785113019775804
```

```python
# 1-Tailed Test [Only +ve Direction]
p_value = 1 - norm.cdf(z_score)
p_value
```

```
0.1192964146582175
```

```python
H0 = "Average session duration <= 5.5 minutes"
HA = "Average session duration > 5.5 minutes " # (one-tailed test)

alpha = 0.05 # Significance Level
if (p_value < alpha):
    print(f"Rejecting the Null Hypothesis H0: {H0}")
    print(f"Accepting the Alternative Hypothesis HA: {HA}")
else:
    print(f"Accepting the Null Hypothesis H0: {H0}")
```

```
Accepting the Null Hypothesis H0: Average session duration <= 5.5 minutes
```

Conclusion: Since p-value (0.119) > alpha (0.05), there is insufficient evidence to suggest the update significantly increased average session duration. The observed increase could be due to random variation.

```python
# One Tailed Test : Teaching Method Example
from scipy.stats import norm
import numpy as np

population_mean = 70
sample_mean = 73
sample_size = 40 # n>=30 : Z-Score
sample_std = 8 # is known to me

Standard_Error = sample_std / np.sqrt(sample_size)
Standard_Error
```

```
1.2649110640673518
```

```python
z_score = (sample_mean - population_mean) / Standard_Error
z_score
```

```
2.3717082451262845
```

```python
# 1-Tailed Test [Only +ve Direction]
p_value = 1 - norm.cdf(z_score)
p_value
```

```
0.008853032903683333
```

```python
H0 = "Average Test Score <= 70"
HA = "Average Test Score > 70: Significant Improvement in Test Score w.r.t Teaching Method"

alpha = 0.05 # Significance Level
if (p_value < alpha):
    print(f"Rejecting the Null Hypothesis H0: {H0}")
    print(f"Accepting the Alternative Hypothesis HA: {HA}")
else:
    print(f"Accepting the Null Hypothesis H0: {H0}")
```

```
Rejecting the Null Hypothesis H0: Average Test Score <= 70
Accepting the Alternative Hypothesis HA: Average Test Score > 70: Significant Improvement in Test Score w.r.t
Teaching Method
```

Result: Since p-value (0.0089) < alpha (0.05), we reject the null hypothesis. There is significant evidence that the new teaching method increased average test scores.

```python
# Two-Tailed Test: Apple Weight Example
from scipy.stats import norm
import numpy as np

population_mean = 150 # in grams
sample_mean = 145
sample_size = 30 # n>=30 : Z-Score
sample_std = 10 # is known to me

Standard_Error = sample_std / np.sqrt(sample_size)
Standard_Error
```

```
1.8257418583505538
```

```python
z_score = (sample_mean - population_mean) / Standard_Error
z_score
```

```
-2.7386127875258306
```

```python
# 2-Tailed Test [Both +ve & -ve Direction]
p_value = 2 * norm.cdf(z_score)
p_value
```

```
0.00616989932054416
```

```python
H0 = "mu = 150 grams (weight equals standard)"
HA = "mu != 150 grams (weight differs from standard)"

alpha = 0.05 # Significance Level
if (p_value < alpha):
    print(f"Rejecting the Null Hypothesis H0: {H0}")
    print(f"Accepting the Alternative Hypothesis HA: {HA}")
else:
    print(f"Accepting the Null Hypothesis H0: {H0}")
```

```
Rejecting the Null Hypothesis H0: mu = 150 grams (weight equals standard)
Accepting the Alternative Hypothesis HA: mu != 150 grams (weight differs from standard)
```

```
"There is significant evidence to suggest that the average weight of the apples is different from 150 grams."
```